

About me

Software Engineer with 10+ years of experience designing, building, and operating large-scale distributed systems. I thrive on the challenge of creating systems that abstract complexity away from the end user, no matter the shape or form, and I particularly enjoy solving non-obvious problems that require deep thinking and creativity.

Work Experience

New Relic (Spain, 100% Remote)

05.2025 – now (1y, 4m)

Cloud-based observability and application performance monitoring platform used by over 85,000 organizations worldwide, ingesting petabytes of telemetry data daily to deliver real-time insights into software, infrastructure, and digital experiences.

Senior Software Engineer, Stream Processing Platform

Own, operate, and build our fully automated Kafka platform infrastructure, running on Kubernetes with over 1,800 brokers distributed across hundreds of clusters.

- Participate in an on-call rotation and bake stability into everything, continually seeking automation opportunities for built-in reliability.

- Led initiative driving \$1.1M/year COGS savings

- Led initiative to apply producer quotas across 140+ shared Kafka clusters

- Led project to build a Kafka Showbacks tool to surface service/team usage and cost attribution across 140+ shared Kafka clusters

Skills: Apache Kafka, Kubernetes, Terraform, Java, Python, Argo CD, New Relic, AWS, Azure, MSK, Strimzi, Apache Flink

Shopify (Spain, 100% Remote)

06.2021 – 08.2024 (3y, 3m)

Leading global commerce company powering over 1.7 million businesses in 175+ countries

Senior Production Engineer, Stream Processing Platform

Led critical streaming infrastructure initiatives and technical projects supporting Shopify's distributed commerce platform.

- Owned and operated Tier 1 (99.9% availability) Apache Kafka clusters serving 29M messages/second at peak during Black Friday/Cyber Monday and Tier 2 (99.95% availability) Apache Flink infrastructure running hundreds of production jobs

- Participated in 24/7 on-call rotation for mission-critical systems, implementing incident response procedures and remediations

- Led capacity planning initiatives, resilience testing (Gamedays), define SLI/SLOs, and disaster recovery exercises to ensure infrastructure readiness for peak traffic events

- Successfully led multiple technical initiatives including KRaft mode migration for Kafka clusters and implementing Flink Kubernetes Operator with Tier 2 requirements

- Established platform engineering solutions enabling hundreds of developers to build reliable data streaming applications

Skills: Apache Flink, Apache Kafka, Kubernetes, Terraform, GCP, Scala, Bash, Observability, Resiliency

Spotahome (Madrid, Spain; 100% remote since March 2020)

09.2017 – 06.2021 (3y, 10m)

Online rental platform operating in 9 countries with 60k+ listings and €300M+ GMV

Manager, Data Analytics & Engineering

05.2018 – 06.2021 (3y, 2m)

Built and led organization of 8 engineers responsible for data infrastructure, analytics, and machine learning capabilities.

- Architected and implemented change data capture (CDC) pipeline to BigQuery, reducing data-to-insight time by orders of magnitude while eliminating daily production failures, built with dbt and scheduled with Airflow (own initiative to re-architect the whole platform)

- Built team from 0 to 8 engineers, creating recruitment process, technical interviews, and onboarding procedures

- Established company-wide data strategy, including technology stack selection, data governance principles, and ML infrastructure

- Led definition and execution of quarterly roadmaps, budget planning, and resource allocation

- Designed and implemented end-to-end data architecture supporting live application features, executive dashboards, and machine learning models

- Collaborated cross-functionally with marketing, product, engineering, and business teams to align data initiatives with company objectives

Skills: Python, SQL, BigQuery, AWS, dbt, Apache Airflow, Kubernetes, Data Analytics, Machine Learning

Data Analytics Engineer

09.2017 – 04.2018 (8m)

Balanced analytical support with data infrastructure development in early-stage data team.

- Led initiative to deploy Apache Airflow on Kubernetes for ETL orchestration
- Implemented ETL architecture leveraging Python and container-based deployment
- Collaborated with product teams on A/B testing and ML research initiatives

Skills: Python, SQL, PostgreSQL, Docker, AWS, Kubernetes, Data Science, Pandas

CST International (Dublin, Ireland)

08.2015 – 07.2017 (2y)

Specialist market research agency focused on employee engagement and brand research

Software Engineer & Data Analyst

Designed and implemented distributed data processing systems and reporting platforms as part of small technical team.

- Architected event-driven data processing system in AWS using serverless components (Lambda, DynamoDB Streams, S3)
- Unified multiple client-based systems into a cohesive platform, improving reliability and reducing maintenance overhead
- Developed web-based reporting systems and survey platforms with comprehensive testing frameworks

Skills: Python, SQL, AWS, DynamoDB, AWS Lambda, Full-stack, Backend, Web2py, MySQL, HTML

UC3M – Charles III University of Madrid (Madrid, Spain)

12.2013 – 06.2014 (7m)

Software Engineer (Research Internship)

Led R&D project in collaboration with Bultaco Motors, an electric motorcycle manufacturer.

- Designed and developed a native Android GPS navigation application based on OsmAnd framework
- Implemented real-time telemetry data collection and display via Bluetooth interfaces
- Developed machine learning system (Artificial Neural Network/MLP) in Java to predict route viability based on historical data
- Created interfaces for information retrieval from SOAP web services

Skills: Java, Android, Machine Learning

Education

UC3M – Charles III University of Madrid (Madrid, Spain)

Telecommunications Engineering (BSc + MSc)

09.2008 – 09.2014

- Top 15% of class with specialization in networking, signal processing, and communications systems
- Coursework: Network protocols (TCP/IP), digital communications, signal processing, programming (Java, Python), electronics
- Final Year Project (Distinction with Honors): *Range Estimation System for Electric Vehicles with Artificial Neural Networks*, available here (in Spanish): <http://dsaiztc.com/PFC.pdf>

Technical Skills

- **Infrastructure & Ops:** Kubernetes, Docker, AWS, GCP, CI/CD, Terraform
- **Networking & Communications:** TCP/IP, distributed systems, real-time messaging systems
- **Streaming Technologies:** Apache Kafka, Apache Flink, Change Data Capture (CDC)
- **Data Systems:** BigQuery, Redshift, PostgreSQL, MySQL, MongoDB
- **Programming:** Proficient in Python, SQL; Have used Java, Scala, Bash, Go
- **Management:** Team building, performance management, cross-functional leadership, strategic planning